[1]:
```
import pandas as pd
```

[24]:
```
df = pd.read_csv("cleaned_dataset (updated).csv")
```

[25]:
```
df.head()
```

[25]:

	Ticket ID	Customer Name	Customer Email	Customer Age	Customer Gender	Product Purchased	Date of Purchase	Ticket Type	Ticket Subject
0	1	Marisa Obrien	carrollallison@example.com	32	Other	GoPro Hero	2021-03-22	Technical issue	Product setup
1	2	Jessica Rios	clarkeashley@example.com	42	Female	LG Smart TV	2021-05-22	Technical issue	Peripheral compatibility
2	3	Christopher Robbins	gonzalestracy@example.com	48	Other	Dell XPS	2020-07-14	Technical issue	Network problem
3	4	Christina Dillon	bradleyolson@example.org	27	Female	Microsoft Office	2020-11-13	Billing inquiry	Account access
4	5	Alexander	bradleymark@example.com	67	Female	Autodesk	2020-02-	Billing	Data loss

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[26]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8469 entries, 0 to 8468
Data columns (total 17 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Ticket ID                     8469 non-null   int64
 1   Customer Name                 8469 non-null   object
 2   Customer Email                8469 non-null   object
 3   Customer Age                  8469 non-null   int64
 4   Customer Gender               8469 non-null   object
 5   Product Purchased             8469 non-null   object
 6   Date of Purchase              8469 non-null   object
 7   Ticket Type                   8469 non-null   object
 8   Ticket Subject                8469 non-null   object
 9   Ticket Status                 8469 non-null   object
 10  Resolution                    2769 non-null   object
 11  Ticket Priority               8469 non-null   object
 12  Ticket Channel                8469 non-null   object
 13  Customer Satisfaction Rating  2769 non-null   float64
 14  First_Response_Minutes        8469 non-null   float64
 15  First_Response_Date           5650 non-null   object
 16  First_Response_Time           5650 non-null   object
dtypes: float64(2), int64(2), object(13)
memory usage: 1.1+ MB
```

[18]:
```
df.columns
```

[18]:
```
Index(['Ticket ID', 'Customer Name', 'Customer Age', 'Customer Gender',
       'Product Purchased', 'Date of Purchase', 'Ticket Type',
       'Ticket Subject', 'Ticket Status', 'Ticket Priority', 'Ticket Channel',
       'Customer Satisfaction Rating', 'First_Response_Minutes',
       'First_Response_Date', 'Average_First_Response_Minutes',
```

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```
      dtype='object')
```

[28]:
```
'Resolution'
```

[28]:
```
'Resolution'
```

[35]:
```
df['Resolution'] = df['Resolution'].fillna(0)
```

[36]:
```
df['Resolution'].isnull().sum()
```

[36]:
```
0
```

[37]:
```
print(df.columns)
```

```
Index(['Ticket ID', 'Customer Name', 'Customer Email', 'Customer Age',
       'Customer Gender', 'Product Purchased', 'Date of Purchase',
       'Ticket Type', 'Ticket Subject', 'Ticket Status', 'Resolution',
       'Ticket Priority', 'Ticket Channel', 'Customer Satisfaction Rating',
       'First_Response_Minutes', 'First_Response_Date', 'First_Response_Time',
       'Resolution_Date'],
      dtype='object')
```

[38]:
```
df.columns = df.columns.str.strip()
```

[39]:
```
df['Resolution'] = pd.to_datetime(df['Resolution'], errors='coerce')
```

[40]:
```
df['Resolution_Date'] = df['Resolution'].dt.date
df['Resolution_Time'] = df['Resolution'].dt.time
```

[41]:
```
df[['Resolution', 'Resolution_Date', 'Resolution_Time']].head()
```

[41]:

	Resolution	Resolution_Date	Resolution_Time
0	1970-01-01	1970-01-01	00:00:00

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1	1970-01-01	1970-01-01	00:00:00
2	1970-01-01	1970-01-01	00:00:00
3	1970-01-01	1970-01-01	00:00:00
4	1970-01-01	1970-01-01	00:00:00

[42]:
```
df[['Resolution', 'Resolution_Date', 'Resolution_Time']].head()
```

[42]:

	Resolution	Resolution_Date	Resolution_Time
0	1970-01-01	1970-01-01	00:00:00
1	1970-01-01	1970-01-01	00:00:00
2	1970-01-01	1970-01-01	00:00:00
3	1970-01-01	1970-01-01	00:00:00
4	1970-01-01	1970-01-01	00:00:00

[43]:
```
df.to_csv("cleaned_supportlytics_dataset.csv", index=False)
```

[44]:
```
df.to_excel("cleaned_supportlytics_dataset.xlsx", index=False)
```

[47]:
```
print(df.columns)
```

```
Index(['Ticket ID', 'Customer Name', 'Customer Email', 'Customer Age',
       'Customer Gender', 'Product Purchased', 'Date of Purchase',
       'Ticket Type', 'Ticket Subject', 'Ticket Status', 'Resolution',
       'Ticket Priority', 'Ticket Channel', 'Customer Satisfaction Rating',
       'First_Response_Minutes', 'First_Response_Date', 'First_Response_Time',
       'Resolution_Date', 'Resolution_Time'],
```

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```
       'First_Response_Minutes', 'First_Response_Date', 'First_Response_Time',
       'Resolution_Date', 'Resolution_Time'],
      dtype='object')
```

[48]:
```
'First_Response_Time'
```

[48]:
```
'First_Response_Time'
```

[49]:
```
df.columns = df.columns.str.strip()
```

[50]:
```
'First Response Time' in df.columns
```

[50]:
```
False
```

[55]:
```
df['First_Response_Time'] = pd.to_datetime(
    df['First_Response_Time'],
    format='%H:%M:%S',
    errors='coerce'
)
```

[56]:
```
df['First_Response_Minutes'] = (
    df['First_Response_Time'].dt.minute +
    df['First_Response_Time'].dt.second / 60
)
```

[57]:
```
df['First_Response_Time'].head(10)
```

[57]:
```
0   2026-02-16 12:15:36
1   2026-02-16 16:45:38
2   2026-02-16 11:14:38
3   2026-02-16 07:29:40
4   2026-02-16 00:12:42
5                   NaT
6                   NaT
```

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```
5                   NaT
6                   NaT
7                   NaT
8   2026-02-16 10:32:47
9   2026-02-16 09:25:48
Name: First_Response_Time, dtype: datetime64[ns]
```

[58]:
```
df['First_Response_Time'] = pd.to_datetime(
    df['First_Response_Time'],
    errors='coerce'
)
```

[59]:
```
df['First_Response_Minutes'] = (
    df['First_Response_Time'].dt.minute +
    df['First_Response_Time'].dt.second / 60
)
```

[60]:
```
df[['First_Response_Time','First_Response_Minutes']].head()
```

[60]:

	First_Response_Time	First_Response_Minutes
0	2026-02-16 12:15:36	15.600000
1	2026-02-16 16:45:38	45.633333
2	2026-02-16 11:14:38	14.633333
3	2026-02-16 07:29:40	29.666667
4	2026-02-16 00:12:42	12.700000

[61]:
```
df = df.drop(columns=['First_Response_Time'])
```

[62]:
```
df.columns
```

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[62]:
```
df.columns
```

[62]:
```
Index(['Ticket ID', 'Customer Name', 'Customer Email', 'Customer Age',
       'Customer Gender', 'Product Purchased', 'Date of Purchase',
       'Ticket Type', 'Ticket Subject', 'Ticket Status', 'Resolution',
       'Ticket Priority', 'Ticket Channel', 'Customer Satisfaction Rating',
       'First_Response_Minutes', 'First_Response_Date', 'Resolution_Date',
       'Resolution_Time'],
      dtype='object')
```

[63]:
```
df.columns = df.columns.str.strip()
```

[64]:
```
df = df.drop(columns=['Customer Email'])
```

[65]:
```
df = df.drop(columns=['Customer Email'], errors='ignore')
```

[66]:
```
df.columns
```

[66]:
```
Index(['Ticket ID', 'Customer Name', 'Customer Age', 'Customer Gender',
       'Product Purchased', 'Date of Purchase', 'Ticket Type',
       'Ticket Subject', 'Ticket Status', 'Resolution', 'Ticket Priority',
       'Ticket Channel', 'Customer Satisfaction Rating',
       'First_Response_Minutes', 'First_Response_Date', 'Resolution_Date',
       'Resolution_Time'],
      dtype='object')
```

[67]:
```
df = df.drop(columns=['Resolution'])
```

[68]:
```
df = df.drop(columns=['Resolution'], errors='ignore')
```

[69]:
```
df.columns
```

[69]:
```
Index(['Ticket ID', 'Customer Name', 'Customer Age', 'Customer Gender',
```

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[69]:
```
Index(['Ticket ID', 'Customer Name', 'Customer Age', 'Customer Gender',
       'Product Purchased', 'Date of Purchase', 'Ticket Type',
       'Ticket Subject', 'Ticket Status', 'Ticket Priority', 'Ticket Channel',
       'Customer Satisfaction Rating', 'First_Response_Minutes',
       'First_Response_Date', 'Resolution_Date', 'Resolution_Time'],
      dtype='object')
```

[70]:
```
df.to_csv("supportlytics_cleaned_dataset.csv", index=False)
```

[71]:
```
df = df.drop(columns=['Resolution_Date', 'Resolution_Time'])
```

[72]:
```
df.columns
```

[72]:
```
Index(['Ticket ID', 'Customer Name', 'Customer Age', 'Customer Gender',
       'Product Purchased', 'Date of Purchase', 'Ticket Type',
       'Ticket Subject', 'Ticket Status', 'Ticket Priority', 'Ticket Channel',
       'Customer Satisfaction Rating', 'First_Response_Minutes',
       'First_Response_Date'],
      dtype='object')
```

[73]:
```
df.to_csv("supportlytics_cleaned_dataset.csv", index=False)
```

[3]:
```
import pandas as pd
```

[5]:
```
import pandas as pd

df = pd.read_csv("supportlytics_cleaned_dataset.csv")
```

[6]:
```
df.head()
```

[6]:

Ticket ID	Customer Name	Customer Age	Customer Gender	Product Purchased	Date of Purchase	Ticket Type	Ticket Subject	Ticket Status	Ticket Priority	Ticke Channe

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	Ticket ID	Customer Name	Customer Age	Customer Gender	Product Purchased	Date of Purchase	Ticket Type	Ticket Subject	Ticket Status	Ticket Priority	Ticke Channe
0	1	Marisa Obrien	32	Other	GoPro Hero	2021-03-22	Technical issue	Product setup	Pending Customer Response	Critical	Socia medi
1	2	Jessica Rios	42	Female	LG Smart TV	2021-05-22	Technical issue	Peripheral compatibility	Pending Customer Response	Critical	Cha
2	3	Christopher Robbins	48	Other	Dell XPS	2020-07-14	Technical issue	Network problem	Closed	Low	Socia medi
3	4	Christina Dillon	27	Female	Microsoft Office	2020-11-13	Billing inquiry	Account access	Closed	Low	Socia medi
4	5	Alexander Carroll	67	Female	Autodesk AutoCAD	2020-02-04	Billing inquiry	Data loss	Closed	Low	Ema

[7]:
```
df['First_Response_Minutes'].mean()
```

[7]:
```
30.228545722713868
```

[8]:
```
df['Average_First_Response_Minutes'] = 30.228545722713868
```

[9]:
```
df[['First_Response_Minutes', 'Average_First_Response_Minutes']].head()
```

[9]:

	First_Response_Minutes	Average_First_Response_Minutes
0	15.600000	30.228546
1	45.633333	30.228546

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[6]:

Date of Purchase	Ticket Type	Ticket Subject	Ticket Status	Ticket Priority	Ticket Channel	Customer Satisfaction Rating	First_Response_Minutes	First_Response_Date
021-03-22	Technical issue	Product setup	Pending Customer Response	Critical	Social media	NaN	15.600000	2023-06-01
021-05-22	Technical issue	Peripheral compatibility	Pending Customer Response	Critical	Chat	NaN	45.633333	2023-06-01
020-07-14	Technical issue	Network problem	Closed	Low	Social media	3.0	14.633333	2023-06-01
020-11-13	Billing inquiry	Account access	Closed	Low	Social media	3.0	29.666667	2023-06-01
020-02-04	Billing inquiry	Data loss	Closed	Low	Email	1.0	12.700000	2023-06-01

[7]:
```
df['First_Response_Minutes'].mean()
```

[7]:
```
30.228545722713868
```

[8]:
```
df['Average_First_Response_Minutes'] = 30.228545722713868
```

[9]:
```
df[['First_Response_Minutes', 'Average_First_Response_Minutes']].head()
```

[9]:

	First_Response_Minutes	Average_First_Response_Minutes
0	15.600000	30.228546
1	45.633333	30.228546

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[8]:
```
df['Average_First_Response_Minutes'] = 30.228545722713868
```

[9]:
```
df[['First_Response_Minutes', 'Average_First_Response_Minutes']].head()
```

[9]:

	First_Response_Minutes	Average_First_Response_Minutes
0	15.600000	30.228546
1	45.633333	30.228546
2	14.633333	30.228546
3	29.666667	30.228546
4	12.700000	30.228546

[10]:
```
df.to_csv("supportlytics_cleaned_dataset.csv", index=False)
```

[12]:
```
df.groupby('Ticket Priority')['First_Response_Minutes'].mean()
```

[12]:
```
Ticket Priority
Critical    29.835665
High        30.222218
Low         30.399390
Medium      30.460466
Name: First_Response_Minutes, dtype: float64
```

[23]:
```
df['Priority_Avg_Response'] = (
    df.groupby('Ticket Priority')['First_Response_Minutes']
    .transform('mean')
)
```

[34]:
```
df[['Customer Age']].head()
```

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